

MOF Design Challenge: group che-4-2

Application: Water purification

Remove larger compounds than water molecules. Hereby, the pore size of the MOFs should be < 2 nm to allow for nanofiltration and only let water and metal ions pass the membrane.

(The kinetic radius, i.e. the effective radius during thermal motion, of water is $2.6 \text{ \AA}$)

<https://pubs.rsc.org/en/content/articlelanding/2023/qm/d3qm00487b>

Stable in water – soft/soft or hard/hard (HSAB theory)

- The complexes become stable when they are constructed from hard acid/bases or soft acid/bases
- Key to obtain high kinetic stability of MOFs in water is by raising the activation energy of hydrolysis. The percentage of effective collisions between water molecules and metal ions of the MOFs will hence be reduced, according to Maxwell-Boltzmann distribution.

[Water-stable metal–organic frameworks \(MOFs\): rational construction and carbon dioxide capture - Chemical Science \(RSC Publishing\) DOI:10.1039/D3SC06076D](#)

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1. What metal node would you choose? Why?

- a. We can either choose a hard acid metal (e.g. Al^{3+} , Zr^{4+} , In^{3+} , Ti^{4+} , Hf^{4+} , Fe^{3+} , and Cr^{3+}) or a soft acid metal (e.g. Zn^{2+} and Cu^{+}). We choose titanium, a hard acid.

2. What type of organic linker? (ditopic / tritopic / functionalized)

- a. To obtain a high-water stability of the MOFs, hard acids should be paired with hard base ligands and vice versa.
- b. Here, hard base ligands can e.g. be carboxylic ligands or another type of ligand with free oxygen atoms which promotes the formation of strong hard acid-O bonds raising the activation energy for hydrolysis.
- c. On the other hand, a soft base, e.g. imidazolyl, triazolyl or tetrazolyl derivatives, can be used in combination with a soft metal.

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To match the linker to the metal node, we choose a hard base linker. To obtain a small pore size to ensure effective water purification, we choose 1,4-benzenedicarboxylate, a small ditopic linker with free oxygen atoms.

Hereby, we have two points to keep in mind:

- Follow the hard/hard and soft/soft HSAB theory to make the MOF water stable
- Use a small and ditropic linker to reduce the pore size.

If we take inspiration from the very water stable UiO-66 MOF with terephthalate (1,4-benzenedicarboxylate) is coordinated to hexameric oxoclusters $[\text{Zr}_6\text{O}_4(\text{OH})_4]$. [Zirconium-Containing Metal–Organic Frameworks \(MOFs\) as Catalysts for Biomass Conversion](#)

This MOF shows a pore size of 6 Å–12 Å (greater than the kinetic radius of water) whereby the pore size preferably should be reduced even further.

https://www.researchgate.net/publication/326598156_METAL-ORGANIC_FRAMEWORKS_APPLIED_FOR_WATER_PURIFICATION

For this reason, a hard metal ion with the same charge (as the cluster assembly and the ligands then can be retained, without altering the charge balance of the MOF), but with a smaller ionic radius could be used. Hereby, if we go up one row in the periodic table, we arrive at Ti^{4+} . This $[\text{Ti}_6\text{O}_4(\text{OH})_4]$ metal cluster seems to could exist in relation to MOFs according to this Korean patent: [KR20190057763A.pdf](#).

Hereby, we propose the MOF with $[\text{Ti}_6\text{O}_4(\text{OH})_4]$ as metal node and 1,4-benzenedicarboxylate as organic linker (isostructural to UiO-66), as a candidate for water purification, as it according to HSAB should be water stable and should have a small pore size.

3. What pore size / property is needed?

- For the goal of nanofiltration, a pore size of <2 nm is needed. Further filtration efforts go all the way down to <1 nm.

4. What challenge might your MOF face

- We do not know if it can exist and is applicable for the specific usage case of water purification.
- Likelihood of rapid nucleation, due to the combination of titanium and 1,4-benzenedicarboxylate as presented in the lecture.

Controlling MOF Growth

1. What problem occurs in uncontrolled nucleation?
 - a. The structure can become amorphous, and the crystal becomes small and uneven whereby the pore size and cavities might vary in size leading to a reduced applicability of the MOF for its intended usage.
2. How do modulators (e.g., acetic acid) help?
 - a. It can slow down the nucleation by temporarily occupying linker binding sites by forming reversible and weak bonds to the metal node, which the ligand with ease can break. Beside this the metal cluster formation is accelerated.
3. Could you design a new modulator?
 - a. Common modulators are monotropic carboxylic acids, like benzoic acid or acetic acid. Most other acids would also work, both amino acids (like proline and glycine) and simple/ diatomic acids (like hydrochloric acid) have seen use as modulators.
 - b. Principally, any monotropic acid could be used as a modulator, if and so far, that it does not interfere with the properties of the linker, but is able to initially compete with it for coordination with the metal nodes.
<https://www.sciencedirect.com/science/article/pii/S0010854521003386>
4. What parameters would you optimize? (temperature, solvent, concentration, etc.)
 - a. As there is a risk that our proposed MOF has a rapid self-assembly, it could be advantageous to lower the temperature during crystallisation and nucleation if the solvothermal method is used as the solvent more slowly would evaporate, so the MOF has more time for ideal framework assembly.
 - b. Another way to control the rapid nucleation, would be to increase the concentration of a modulator. Because we have chosen 1,4-benzenedicarboxylate for our proposed MOF, a natural choice of modulator is benzoic acid.